Software Test Plan (STP)

Project: Task Manager Application

Version: 1.0

Section: J

Authors:

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Status: Finalized (Test Execution Scheduled)

Revision History

Version	Date	Author	Change Summary
1.0	15-09-2026	Student Team	Initial Software Test Plan created for the Task Manager Application based on the SRS and SAD documents.

Approvals

Role	Name	Signature / Date
Course Coordinator		

1. Introduction

Purpose

This document defines the test plan for the Task Manager Application, Version 1.0. It outlines the objectives, scope, strategy, resources, schedule, and responsibilities for testing the application. It is based on the Software Requirements Specification (SRS) and the Software Architecture and Design (SAD) documents prepared for this project.

Scope

Testing covers the core features of the Task Manager Application, including user registration and login, task creation, editing, deletion, and completion, priority and status management, categories and tags, due dates and reminders, search and filtering, progress summaries, notifications, user preferences, and administrator functions where enabled. Testing also covers the non functional and security requirements defined in the SRS, such as performance, reliability, data integrity, accessibility, and protection of user credentials and task data. Third party notification services and any external calendar or synchronization APIs are tested only at the interface level and not internally, since their implementation is outside the scope of this project.

References

- Software Requirements Specification (SRS), Task Manager Application, Version 1.0, dated 09-09-2026
- Software Architecture and Design (SAD) Specification, Task Manager Application, Version 1.0, dated 14-09-2026
- Requirements Traceability Matrix (RTM) contained in the SRS and SAD documents

Definitions

Task, a unit of work to be completed. User, an authenticated application user. Priority, importance level assigned to a task. Status, current state of a task. Due Date, date or time by which a task should be completed. Reminder, a notification associated with a task. UI, User Interface. API, Application Programming Interface. CRUD, Create, Read, Update and Delete. SRS, Software Requirements Specification. SAD, Software Architecture and Design. RTM, Requirements Traceability Matrix.

2. Test Items

- Authentication Module (registration, login, logout)
- Task Module (create, edit, delete, complete or reopen tasks)
- Access Control (task ownership and authorization checks)
- Scheduling Module (priority, status, categories or tags, due dates, reminders)
- Search Module (search, filter, sort)
- Dashboard Module (progress summaries)
- Notification Module (reminders, notification preferences)
- Administration Module (user and account management, where enabled)
- Application Database (persistence of users, tasks, categories, reminders, preferences, and task history)

3. Features to be Tested

Features mapped to SRS requirement IDs:

- TMA-F-001 to TMA-F-004: User registration, login, invalid login handling, and logout
- TMA-F-010 to TMA-F-014: Task creation, editing, deletion, completion or reopening, and ownership based access control
- TMA-F-020 to TMA-F-024: Task priority, status, categories or tags, due dates, and reminders
- TMA-F-030 to TMA-F-033: Task search, filtering, sorting, and progress summary
- TMA-F-040 to TMA-F-042: Notifications, notification preferences, and administrator user management
- TMA-NF-001: Dashboard response time within 3 seconds for 95 percent of requests
- TMA-NF-002: 99.5 percent monthly availability excluding scheduled maintenance
- TMA-NF-003: Protection of passwords and sensitive user or task data
- TMA-NF-004: Data consistency across create, update, complete, and delete operations
- TMA-NF-005: Responsive and accessible user interface on supported screen sizes
- TMA-SR-001 to TMA-SR-005: HTTPS or TLS usage, password hashing, authorization, session protection, and security event logging

4. Features Not to be Tested

- Internal implementation of third party notification services, since only the interface with the Notification Module is tested
- Internal implementation of any external calendar or synchronization API, if integrated in a later phase
- Underlying database engine or hosting infrastructure internals, which are assumed to function as documented by the provider
- Browser or operating system level defects that are not caused by the application itself

5. Test Approach / Strategy

Levels

- Unit tests at the module level, for example the Authentication Module and Task Module
- Integration tests between modules, for example the Task Module and the Application Database, and the Notification Module and the Scheduling Module
- System tests covering end to end task management flows from login to task completion
- Acceptance tests (UAT) validating the application against the SRS acceptance criteria

Types

- Functional testing of all core features listed in Section 3
- Regression testing after each change to confirm existing features remain unaffected
- Performance testing of dashboard response time and search or filter operations under normal load
- Usability and accessibility testing of the UI on supported desktop and mobile screen sizes
- Security testing covering authentication, authorization, session handling, and data protection

Entry Criteria: A stable build of the Task Manager Application is delivered, required test data such as sample user accounts and tasks is available, and the test environment is ready.

Exit Criteria: All planned test cases are executed, no critical or high severity defects remain open, and all acceptance criteria defined in the SRS are satisfied.

5.1 Security Validation

- Verify passwords are stored only as secure hashes and never in plaintext (TMA-SR-002, TMA-NF-003)
- Verify HTTPS or TLS is enforced for all client server communication (TMA-SR-001)
- Verify users cannot access or modify tasks belonging to another user (TMA-F-014, TMA-SR-003)
- Verify session tokens are protected and invalidated on logout or expiry (TMA-SR-004)
- Verify security relevant events such as failed logins are logged without recording passwords (TMA-SR-005)
- Basic input validation and fuzz style checks on login and task entry fields

6. Test Environment

Hardware: Standard desktop, laptop, and mobile devices with supported input and display capability, as no specialized hardware is required.

Software: Task Manager Application build under test, supported web browser or mobile or desktop client, backend application server, and application database.

Tools: A UI test automation tool (for example Selenium or an equivalent) for functional and regression testing, an API testing tool (for example Postman) for backend endpoints, a load testing tool (for example JMeter or an equivalent) for performance testing, and an issue tracker (for example Jira or an equivalent) for defect tracking.

Test Data: Sample user accounts, sample tasks with varying priorities, statuses, categories, and due dates, and sample reminder configurations.

7. Test Schedule

Milestones:

- Test case design: 18-Sep-2026
- Environment setup: 20-Sep-2026
- Test execution start: 22-Sep-2026
- Test execution end: 29-Sep-2026
- UAT: 30-Sep-2026 to 02-Oct-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases, both manual and automated
- Test Scripts
- Test Data
- Test Execution Logs

- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Tharunyah Sudhakar	Prepare the test plan and coordinate test execution
Test Engineer	Yajat Mathur	Design and execute test cases, log defects
Test Engineer	Vismay Chandra Dev	Design and execute test cases, log defects
Developer	Vishnu Narayanan Ajay	Support defect fixes and triage
Product Owner	Course Coordinator	Approve test results and sign off on readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early builds from the development team for smoke testing
Test environment downtime	Maintain a backup environment or a local setup that mirrors production
Dependency on a third party notification service	Engage the service early and use test stubs where the real service is unavailable
Incomplete or inconsistent test data	Prepare and validate sample user and task data before execution begins

11. Assumptions & Dependencies

- The application database will be stable and available throughout the test cycle
- Test data such as sample accounts and tasks will be provided or created before execution begins
- The notification service, where used, will be available for interface level testing
- Administrator features will be enabled in the test environment where applicable

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable for more than 4 hours
- The build is too unstable and blocks more than 30 percent of planned test cases

Resume testing if:

- Blocking defects are resolved
- The environment is stabilized

13. Test Case Management & Traceability

The Requirements Traceability Matrix (RTM) in the SRS and SAD documents ensures mapping of requirements to test cases. Examples:

- TMA-F-001 (User registration) maps to TC-Auth-01
- TMA-F-010 (Create task) maps to TC-Task-01
- TMA-F-014 (Task access authorization) maps to TC-Sec-01
- TMA-NF-001 (Dashboard response time) maps to TC-Perf-01

- TMA-SR-002 (Password hashing) maps to TC-Sec-02

13.1 Sample Test Cases

Test Case ID	Requirement ID	Description	Expected Result
TC-Auth-01	TMA-F-001	Register a new user with valid required information	A unique user account is created
TC-Auth-02	TMA-F-002	Log in with valid credentials	User is authenticated and reaches the task dashboard
TC-Auth-03	TMA-F-003	Log in with invalid credentials	Login is rejected with an appropriate error message
TC-Task-01	TMA-F-010	Create a task with a title and optional description	The task is saved and appears in the task list
TC-Task-03	TMA-F-012	Delete a task after confirmation	The task is removed from active task views
TC-Sec-01	TMA-F-014	Attempt to access another user's task	Access is denied
TC-Org-04	TMA-F-023	Set or change a task due date	The due date is stored and displayed correctly
TC-Search-01	TMA-F-030	Search tasks by title or description keyword	Matching tasks are returned and non matching tasks are excluded
TC-Perf-01	TMA-NF-001	Load the dashboard under normal load	95th percentile response time is 3 seconds or less
TC-Sec-02	TMA-SR-002	Inspect stored password data	Only secure password hashes are present, no plaintext passwords

14. Test Metrics & Reporting

Metrics collected:

- Percentage of test cases executed
- Percentage of test cases passed or failed
- Defect density
- Defect aging
- Requirement coverage

Reports:

- Daily execution status
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Tharunyah Sudhakar	Tharunyah Sudhakar
Dev Lead	Vishnu Narayanan Ajay	Vishnu Narayanan Ajay
Product Owner	Course Coordinator	Course Coordinator